

# The Legacy of the Opioid Epidemic

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## Abstract

In this paper, we examine how the opioid epidemic affected local population growth and migration decisions in the United States. Leveraging variation in local exposure to the epidemic—driven by Purdue Pharma’s targeted marketing of OxyContin and proxied by 1996 cancer mortality rates—we find that commuting zones with greater exposure experienced significant population declines, concentrated among individuals aged 18 to 44. Direct mortality from drug-induced deaths did not contribute substantially to this decline. Instead, the reduction was driven by changes in migration patterns: exposure prompted the selective out-migration of college-educated women. Our findings indicate that the epidemic reshaped local population composition and contributed to the long-run demographic divergence across commuting zones.

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# I. Introduction

The opioid epidemic is one of the most devastating and far-reaching public health crises in the history of the United States. It has driven increases in mortality, widespread addiction, economic hardship, and shifts in political preferences (Maclean et al., 2020; Arteaga and Barone, 2025). Over the past three decades, the crisis has evolved from the misuse of prescription opioids to the growing prevalence and use of heroin and fentanyl, leaving a profound impact on the families and communities most directly affected. The crisis has unfolded alongside a broader decline of income convergence among regions (Ganong and Shoag, 2017, Moretti, 2011). This divide is also reflected in patterns of population growth. Local population decline has become a large and growing challenge: more than half of all counties lost population between the 2010 and 2020 censuses (Mackun et al., 2021).

In this paper, we examine how the opioid epidemic and the economic hardship it induced altered local population growth and internal migration—two demographic processes with important implications for social mobility and economic development. To isolate the effects of the epidemic from confounding socioeconomic trends, we exploit quasi-exogenous variation in local exposure to Purdue Pharma’s 1996 marketing campaign for OxyContin. Our identification strategy leverages detailed features of this campaign, based on unsealed court records from litigation against Purdue Pharma—the manufacturer of OxyContin, the prescription opioid at the center of the crisis. Those records show that at the dawn of the opioid epidemic in 1996, pharmaceutical marketing efforts were concentrated in the cancer pain market with a plan to quickly expand to the much larger non-cancer pain market in those *same* geographic areas. This targeting implied that noncancer patients in high-cancer areas were disproportionately exposed to the opioid epidemic and the unfortunate chain of events that followed. As in Arteaga and Barone (2025), we use cancer mortality in 1996—i.e., before the unfolding of the epidemic—as a measure of exposure. This work and subsequent research, show that mid-nineties cancer mortality is a strong indicator of exogenous exposure to the opioid epidemic.<sup>1</sup>

Our empirical analysis combines data from several sources. We use the Decennial Censuses and the American Community Survey (ACS) to construct local population counts and, jointly with IRS Statistics of Income (SOI) Tax Stats data, to measure of migration behavior. To capture the economic impact of the epidemic, we rely on data from the Longitudinal Employer–Household Dynamics (LEHD) program, which provides detailed county-level information on employment and earnings. We complement these with data on Supplemental Nutrition Assistance Program (SNAP) receipt rates and applications

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<sup>1</sup>This strategy has been used in the literature to study its consequences on health and economic outcomes, see Buckles et al. (2022); Cohle and Ortega (2023); Olvera et al. (2023); Siegal (2023), among others.

from the Food and Nutrition Service of the U.S. Department of Agriculture. Finally, we use data from the National Vital Statistics System (NVSS) to construct fertility outcomes, opioid-related mortality rates, and cancer mortality. Our analysis focuses on 583 commuting zones with populations exceeding 25,000 in 1996, observed annually from 1980 to 2020, resulting in a panel dataset comprising 17,490 CZ-year observations.<sup>2</sup>

While the opioid epidemic stands as one of the most tragic public health crises of the past century—marked by excess drug-related mortality in the communities most affected—it also contributed to a deterioration of local economic conditions. We extend prior literature by documenting that exposure to the epidemic led to lower employment rates and earnings, with this economic distress being gendered.<sup>3</sup> In particular, we estimate that exposure to the opioid epidemic generated greater economic hardship for women. We find a continuous decline in the employment-to-population ratio among women without a college degree and a reduction in average earnings among women with a college degree. We complement these results by examining participation in the SNAP program—the largest federal antipoverty program in the US. We find a strong link between opioid exposure and rising SNAP enrollment, exposure to the epidemic predicts increased applications among both male and female applicants, with stronger effects for women.

The heightened mortality and economic stress brought about by the epidemic raise the question of how places and people have adjusted to this shock. We focus on local population growth and composition and interpret these changes through three key demographic margins: migration, deaths, and births.

The opioid epidemic has contributed to regional divergence among US communities. Our findings reveal that exposure to the opioid epidemic led to significant *relative* population declines 10 and 20 years after the onset of the crisis. We estimate that by 2010, areas more exposed to the epidemic experienced 1.2 percentage points lower population growth per one-standard-deviation increase in exposure; by 2020, this effect was 2.5 percentage points. To put this number into context, between 2000 and 2020 the average local population change was 8.6%, our estimates suggest that one standard deviation of epidemic exposure predicts a population decline of about 2.5%, or 28% of the observed average change (0.025/0.086).<sup>4</sup> These effects are stronger among individuals aged 18 to 44 years old and persist over time. By 2020, the young adult population declined by 4.1 p.p. for a one standard deviation of exposure—or 20% of the inter-quartile local population change. Shifts in the age distribution of these places were accompanied by increases

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<sup>2</sup>These commuting zones account for over 97 percent of the total population in 1996. The availability and sparsity of the data vary with the outcome of interest, section III. provides additional details on data availability and sources.

<sup>3</sup>Add some cites to prior work.

<sup>4</sup>For further reference, [Arteaga and Barone \(2025\)](#) estimate that areas with a one-standard-deviation higher cancer mortality rate in 1996 experienced an additional 0.97 opioid doses prescribed per capita by 2012—the year when prescription rates peaked—equivalent to 65% more than the baseline mean. By 2017, this translated into a 46% increase in drug-related mortality relative to its pre-epidemic average.

in the sex ratio—i.e., women become relative more scarce in high exposure places.

To shed light on how the opioid epidemic reshaped local populations, we turn to the underlying demographic components of population change. Consistent with models predicting that migration is a primary channel of adjustment to local economic shocks, we find that exposure to the epidemic induced significant migratory responses. A one-standard-deviation increase in exposure led to a 4% increase in out-migration rates. In line with the gendered economic effects, these responses are driven by young women, for whom we estimate a 12% increase in out-migration rates. The effects are strongest among women with some college education, while we find no statistically significant migration responses for men. Back-of-the-envelope calculations suggest that these migratory responses account for most of the observed population changes between 2000 and 2020. We examine the degree of directness in the migration response—that is, the extent to which movers from highly exposed areas targeted less exposed destinations. On average, we find that movers relocated to better areas—measure as having lower exposure to the epidemic—although this targeting was not perfectly directed. Individuals from more exposed origins tended to move to less exposed destinations, but not to the least exposed ones.

In 2018, the drug-related mortality rate for men was 2.5 times higher than that for women. However, when examining the effects of exposure to the opioid epidemic, we find that the effect sizes are broadly similar across sexes. We propose a simple counterfactual exercise to understand whether these mortality impacts are large enough to contribute to the relative population decline. We compute counterfactual local population counts by adding back the accumulated deaths from drug-induced causes to the observed population counts. We then compare the evolution of the population. Among those aged 18 to 44—the group most affected by the epidemic—we find that population growth would have been 0.25% higher. This is an order of magnitude smaller than the estimated changes induced by the epidemic, suggesting that the mortality channel played only a marginal role in slowing population growth.

Heterogeneous fertility responses may generate a demographic counterforce, allowing certain places to experience population growth even in the presence of net out-migration. Our findings reveal that exposure to the opioid epidemic led to an increase in fertility rates. We estimate that by 2018, a one-standard deviation increase in exposure raised fertility rates at the commuting zone level by approximately 5.5%. This increase is primarily driven by unmarried women, for whom we detect rising fertility beginning in 2006, with effects that persist and grow over time. Among married women, we estimate fertility increases that are about half as large and statistically significant starting in 2013. The effects are concentrated among women in their late twenties and are higher for those without a college degree. The timing of these results implied that they have little explanatory power in changes in local population. Nonetheless, this channel has important

implications for the long run. These changes in fertility contribute to the widening divide in family formation patterns by education level (McLanahan, 2004; Lundberg and Pollak, 2014; Kearney, 2022).

This paper contributes to three distinct literatures. First, we contribute to the literature on the effects of local economic shocks and the adjustments of workers to these shocks. This literature has documented how trade shocks (Autor and Dorn, 2013; Greenland et al., 2019), robots adoption (Acemoglu and Restrepo, 2020), the fracking boom (Wilson, 2022), and the Great Recession (Yagan, 2019) have shaped economic conditions of places and how workers have (or not) responded to these shocks. We contribute to these works by showing how the opioid epidemic shaped local population growth and migration. In particular, we document selective migratory responses that vary systematically by demographic and socioeconomic characteristics and stand in contrast to prior work that has found small or insignificant migration responses to negative economic shocks (Autor and Dorn, 2013; Yagan, 2019; Borusyak et al., 2022). Our findings underscore the importance of measuring migration using origin-destination flow pairs disaggregated by group, rather than relying on net migration flows, which may mask meaningful heterogeneity in population responses.

We also add to the literature on the socioeconomic determinants of family formation and fertility. A large body of work has documented how economic conditions and individual socioeconomic status shape decisions around fertility. Foundational work by Becker (1960, 1973) emphasized the role of income, opportunity costs, and specialization in determining family behaviors. Subsequent research has shown that economic insecurity—particularly limited labor market prospects for less-educated men—increases nonmarital fertility (Wilson, 1987, 1996; Black et al., 2005; Autor et al., 2019). This literature has also examined the family formation effects of positive economic shocks to men such as the fracking and coal booms (Black et al., 2013; Kearney and Wilson, 2018), finding increases in fertility rates with no meaningful changes in the marriage. We complement this work by showing that this major public health crisis and its economic consequences have contributed to divergent patterns of childbearing.

Lastly, we extend the literature on the effects of the opioid epidemic by estimating its impact on fertility and migration. Prior research has documented the epidemic’s effects on mortality, addiction, disability claims, municipal finances, housing prices, children’s outcomes, and political behavior (Buckles et al., 2022; Park and Powell, 2021; Cornaggia et al., 2022; Arteaga and Barone, 2025). See Maclean et al. (2020) for a review.

## II. Background: The Unfolding of the Opioid Epidemic

The United States has experienced an unprecedented crisis related to the misuse of and addiction to opioids. As of 2022, over 700,000 lives had been lost to opioid overdoses

(CDC, 2023). During the last decade, a sizeable body of research has studied the origins of the opioid crisis and the factors that shaped its evolution and propagation. This literature has established that the pharmaceutical industry and healthcare providers played a critical role in the origins of the crisis (Eichmeyer and Zhang, 2020; Miloucheva, 2021; Alpert et al., 2022; Arteaga and Barone, 2025). In particular, the aggressive and deceptive marketing of potent opioids with high potential for addiction directed toward physicians, in a setting with financial incentives for doctors to increase prescriptions and with weak monitoring, created the perfect platform for the crisis to unfold.

The beginning of the opioid epidemic is traced to the introduction of OxyContin to the market in 1996 (Quinones, 2015). OxyContin is a prescription opioid manufactured by Purdue Pharma that changed the standard of practice for the treatment of noncancer and nonterminal pain. Prior to the mid-1990s, pain management had focused on cancer and end-of-life pain treatment because of care providers' fears of the risk of severe addiction (Melzack, 1990). Soon after the introduction of OxyContin, reports of abuse and addiction started to be covered by the media. By 2001, West Virginia, Virginia, Ohio, and Kentucky were pursuing class-action lawsuits against Purdue Pharma and other pharmaceutical companies. During this time, United States Attorneys for Maine and Virginia, physicians, and community leaders raised concerns to the FDA and Congress about the level of OxyContin abuse (Meier, 2018). The prescription of opioids continued to increase during the 2000s with limited restrictions. At their peak in 2012, opioid prescriptions numbered 81.3 prescriptions per 100 persons (CDC, 2020). The rate of substance use disorder grew by a factor of six between 1999 and 2009 (Paulozzi et al., 2011), and prescription opioid mortality grew by a factor of five (Maclean et al., 2020).

In response to the widespread misuse of prescription opioids and in particular OxyContin, in 2010, Purdue Pharma introduced an abuse deterrent formulation of OxyContin. Evans et al. (2019) and Alpert et al. (2018) show that the reformulation unfortunately led many consumers to substitute toward a dangerous and inexpensive alternative: heroin. As a result, deaths, poisonings, emergency room visits, and enrollments in treatment programs for heroin abuse increased. In particular, between 2010 and 2013, heroin death rates increased by a factor of four with no reduction in the combined heroin and opioid death rate (Evans et al., 2019). From 2013 to this day, the epidemic has been characterized by surging deaths related to the use of synthetic opioids, particularly fentanyl. Fentanyl, an extremely potent synthetic opioid, is more profitable to manufacture and distribute than heroin and has a higher risk of overdose. Indeed, fentanyl-related deaths account for almost the entire increase in drug overdose mortality between 2014 and 2021.

Mortality from opioids is only one of the many social costs associated with the opioid epidemic. Some 10.1 million people aged 12 or older are estimated to have misused opioids in the past year in the U.S. (SAMHSA, 2020). These numbers are orders of magnitude larger than the number of deaths and help rationalize why the opioid epidemic

has disrupted health and economic opportunities, affecting individuals and their communities. Notably, the epidemic has increased disability rates and Supplemental Nutrition Assistance Program (SNAP) utilization (Powell et al., 2020; Savych et al., 2019; Arteaga and Barone, 2025) and affected labor force participation and employment rates (Krueger, 2017; Ouimet et al., 2020).

This evidence highlights the profound toll of the opioid epidemic on individuals, families, and communities. While its health and economic consequences are well documented, much less is known about how the epidemic has shaped key demographic behaviors. In the rest of the paper, we examine how exposure to the opioid crisis affected population growth and internal migration decisions—two critical processes that influence long-run community structure, social mobility, and economic development.

### III. Data

**Mortality.** We use restricted-use data from the Detailed Multiple Cause of Death (DMCD) files from 1989 to 2018 to construct mortality measures. These files record every death in the United States, including the county of residence, the underlying cause of death, and up to 20 additional contributing causes, and thus represent a census of deaths in the U.S. The 1989–1998 data use ICD-9 codes to classify causes of death, while the 1999–2018 data use ICD-10 codes. We compute the 1996 cancer mortality rate to measure exposure to the opioid crisis.<sup>5</sup> Figure 1 shows the distribution of cancer mortality across geographies in 1996 and Table 1 shows summary statistics for cancer-mortality in the pre- and post-periods.

We construct a measure of drug-related overdose deaths to study the mortality effects of the epidemic. We define drug-induced mortality to include deaths from poisoning and medical conditions caused by the use of legal or illegal drugs, including prescription opioids, heroin, and synthetic opioids such as fentanyl.<sup>6</sup> We analyze drug-related deaths separately for women and men, and also leverage educational attainment data included in death certificates to document effects by education level—specifically, for individuals with some college or more, and those without any college education.<sup>7</sup> Panel (b) of Figure 1 presents the geographic distribution of the evolution of drug-induced mortality between 2018 and 1996. Table 1 presents summary statistics for drug-induced mortality overall

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<sup>5</sup>We define cancer deaths as those caused by malignant neoplasms (codes 140–208 in ICD-9 and C00–C97 in ICD-10) and in situ neoplasms, benign neoplasms, and neoplasms of uncertain or unknown behavior (codes 210–239 in ICD-9 and D00–D48 in ICD-10).

<sup>6</sup>Drug-related overdose deaths can be difficult to categorize; for this reason, we favor a broad measure of mortality. This category includes codes E850–E858, E950.0–E950.5, E962.0, and E980.0–E980.5 in ICD-9; and X40–X44, X60–X64, X85, Y10–Y14, and T36–T39, T40–T50 in ICD-10.

<sup>7</sup>The 2018 data were the first for which educational attainment was consistently reported using the 2003 revision of the death certificate across all deaths (unless missing entirely). In prior years, some deaths still used the 1989 revision’s categorization. This change prevents us from constructing more granular education categories. The next sub-section provides more detail on these definitions.



and by sex. Mortality rates for men are twice as high as those of women before the start of the epidemic. As the epidemic unfolded, mortality rates for both sexes increased at similar rates.

**Population counts.** We use three data sources for population counts. First, the Survey of Epidemiology and End Results (SEER), which reports population at the county level and by age, race, sex, and Hispanic origin. We use these data to construct population counts for decennial census years and denominators for mortality rates, e.g., drug-induced mortality by sex. SEER does not provide population counts by educational attainment. For education-specific outcomes, we leverage data from the Census and the ACS. These data are at the county level and include the years 1990 and 2000 from Census and 2006 onward from the ACS. When using Census and ACS data to construct denominators, we linearly interpolate population counts.

**Economic outcomes.** We collect data from the Longitudinal Employer-Household Dynamics (LEHD) program, which provides detailed information on employment and earnings at the county-level and by age, race, sex and education. Specifically, we use the Quarterly Workforce Indicators (QWI) employment counts and average monthly earnings. We construct sex-specific employment-to-population ratios as the number of stable jobs divided by the working age population. We also construct a measure of average earnings by sex as the ratio between earnings across all permanent jobs to the total number of permanent jobs.

To supplement our analysis on economic hardship we leverage data on SNAP benefit receipt rates and applications from the Food and Nutrition Service of the Department of Agriculture. We use data on county-level participation in the month of January for all years spanning 1989–2020, focusing on beneficiaries of food stamps (FSP) and electronic benefit transfer (EBT) in the context of the program. We then aggregate the county-level counts to compute the share of beneficiaries in the population at the CZ level.<sup>8</sup> These data allow us to test for differential trends before the epidemic but do not allow for the estimation of sex-specific impacts. To do so, we use the Quality Control (QC) database which contains detailed information on the economic and demographic circumstances of a sample of households selected for review as part of the SNAP QC System.

**Migration.** We collect data on county-to-county migration flows from the IRS Statistics of Income (SOI) Tax Stats. These data are based on the universe of Forms 1040 filed and processed by the IRS during a calendar year and are available for filing years 1990 through 2021. The dataset includes the number of personal exemptions claimed, which approximates the number of individuals in a given county. Migration flows are identified based on year-to-year address changes reported on individual income tax returns. These data

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<sup>8</sup>When information at the local level is not available, we impute the state-level share of SNAP recipients.



capture migration patterns across all U.S. counties and include both inflows—the number of new residents moving into a county and their origin—and outflows—the number of residents leaving a county and their destination. While these data cover most internal migration, they do not include demographic information. To examine migration flows for specific subpopulations, we complement the IRS data with information from the Census and the ACS. Specifically, we construct the number of moves from each geographic area based on individuals’ reported locations in the previous year and rely on person weights to generate representative estimates.<sup>9</sup> Informed by Foschi et al. (2023), we drop the four least populous states of the contiguous United States (South and North Dakota, Vermont, and Wyoming) from our analysis sample due to the volatile state population swings observed in the ACS data.

**Fertility.** We use the National Center for Health Statistics (NCHS) Natality and Linked Births files to construct fertility rates. These data represent the universe of births in the United States and include geographic identifiers—such as the county of birth—as well as detailed demographic characteristics of the birthing individual, including age, marital status, and educational attainment. From these data, we compute average fertility rates at the commuting-zone level. The overall fertility rate is defined as the number of births to individuals in a given age group divided by the population in that age group.

We further disaggregate fertility rates by marital status and educational attainment.<sup>10</sup> For the analysis by marital status, we define the married (unmarried) fertility rate as the number of births to married (unmarried) individuals in a given age group divided by the population of married (unmarried) women in that age group. The numerator is obtained directly from the NCHS Files; the denominator is constructed using data from the U.S. Census and the American Community Survey (ACS).

To compute fertility rates by educational attainment, we classify mothers into two groups: those with some college or more, and those without any college education. The NCHS data include information on years of schooling, but we reclassify education levels to account for changes in how this information is reported across time and states. We define “some college or more” as having completed more than 13 years of education, any number of years of college without a degree, or any higher degree (e.g., a bachelor’s or master’s degree). If education is reported but does not meet this definition, we classify the individual as having “no college education.” As with the marital status analysis, the denominator is constructed using Census and ACS data.

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<sup>9</sup>Unfortunately, the 1990 and 2000 Censuses ask about residence five years prior. Therefore, for Census years, we construct the number of moves based on this retrospective question.

<sup>10</sup>The Natality Detail Files include information obtained from each U.S. birth certificate. There is some variation across states and over time in the availability of specific variables. Table A.xx documents the variables used and reports missingness rates by state. In some cases, we exclude states from specific analyses due to data limitations; we note these exclusions when relevant.

**Summary statistics.** Table 1 presents summary statistics.

**Geographic harmonization.** We use the crosswalks developed by Autor and Dorn (2013) to aggregate all data to CZ level. Some CZs cross state borders; when this happens, the CZ is assigned to the state with the larger share of the CZ population. This criterion helps preserve the strong within-cluster and weak between-cluster commuting ties.

In sum, we build a dataset at the commuting-zone level, covering the period from 1989 to 2018 for our outcome variables and the measure of exposure to the epidemic. We choose commuting zones as our unit of observation since it is the geographic space that captures one’s economic life—which often spans beyond county borders. We restrict our sample to areas with more than 25,000 residents. This represents 97% of the total population. Our final dataset is a balanced panel of 583 commuting zones and consists of over 17,490 observations. Table 1 presents summary statistics for key variables for our analysis sample, in pre- and post-periods.

## IV. Empirical Strategy

To identify the effect of the opioid epidemic on local labor markets, population and demographics of places, we exploit rich geographic variation in the early promotional efforts of prescription opioids as an exogenous source of exposure. Following the approach in Arteaga and Barone (2025), we use 1996 cancer mortality as a proxy for local exposure to the opioid epidemic. The rationale behind this strategy is that the size of the cancer pain market in the mid-1990s was a key determinant of where pharmaceutical companies initially concentrated their marketing efforts for opioids. A large body of research—and multiple legal rulings—has established a strong link between these early promotional campaigns and the subsequent unfolding of the opioid crisis.<sup>11</sup>

Our empirical strategy interacts cancer mortality rates with year dummies in a continuous-treatment event-study design:

$$\Delta y_{ct} = \sum_{\tau=1989}^{2018} \phi_{\tau} CancerMR_{c,1996} \mathbf{1}(Year = \tau) + \sum_{\tau=1989}^{2018} \alpha_{\tau} X_{ct_0} \mathbf{1}(Year = \tau) + \gamma_{st} + v_{ct}, \quad (1)$$

where  $c$  indexes the commuting zone,  $s$  the state, and  $t$  the year.  $t_0$  corresponds to 1996—the year of OxyContin’s launch—or the closest year with available data; e.g., 2000 when Census data are used. We define  $\Delta$  as the long-change operator: for any random variable  $W_{ct}$ ,  $\Delta W_{ct} = W_{ct} - W_{ct_0}$ .

The outcome of interest is given by  $y_{ct}$ .  $X_{ct_0}$  is a vector of baseline demographic variables such as the share female, White, and age groups.  $CancerMR_{c,1996}$  is the cancer mortality rate in commuting zone  $c$  in 1996 and it is interacted with a full set of year fixed

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<sup>11</sup>Appendix B. presents additional empirical evidence probing the validity of this empirical approach.

effects indexed by  $\tau$ . In this specification, the coefficients for the pre-epidemic period; i.e.,  $\phi_{1989}$ ,  $\phi_{1990}$ , to  $\phi_{1995}$ , test whether outcomes in higher and lower cancer mortality areas followed similar trends before 1996. In the result section, we present scaled estimates of the  $\phi_\tau$  coefficients, we refer to these as *effect sizes*. Specifically, we multiply the estimated coefficient by the standard deviation of 1996 cancer mortality rates and divide by the mean of the outcome variable in the baseline year  $t_0$ . This yields the percent change in the outcome associated with a one standard deviation increase in exposure.

This research design allows us to control for state-specific trends and state-level policy changes which were common during this period, as well as time invariant commuting zone characteristics. When estimated on a perfectly balanced panel, this specification is equivalent to a regression in levels with commuting zone fixed effects. However, our panel is not perfectly balanced for some outcomes—e.g., fertility by marital status cannot be constructed for California after 2016. In such cases, the specification in Equation (1) is more efficient, and we adopt it as our baseline approach.

We also consider a more parsimonious specification that aggregates effects for time periods but still permits some heterogeneity. We define four time bins [1990-1995], [1996 - 2000], [2001- 2010], and [2011 - 2018]:

$$\begin{aligned} \Delta y_{ct} = & \phi_{95} CancerMR_{c,1996} D_{[90-95]}(t) + \phi_{10} CancerMR_{c,1996} D_{[01-10]}(t) \\ & + \phi_{18} CancerMR_{c,1996} D_{[11-18]}(t) + \sum_g \alpha_g X_{ct0} D_g(t) + \gamma_{st} + \nu_{ct}, \end{aligned} \quad (2)$$

where  $D_g(t) = \mathbf{1}\{t \in g\}$  is the indicator for time bin  $g$ , and the excluded category is 1996-2000 to ease comparison with the rest of the analysis. The rest of the variables are defined as in Equation (1).

Local population growth is persistent over time (see [Greenland et al., 2019](#), among others), ignoring this persistence would introduce bias in the estimation of the effects of the opioid epidemic on local population growth.<sup>12</sup> To estimate the impact of the opioid epidemic on local population growth we consider the following equation:

$$\Delta \ln(pop_{c,t}) = \phi CancerMR_{c,1996} + \beta \Delta \ln(pop_{c,t-10}) + \alpha X_{c,1990} + \gamma_s + \epsilon_{ct}, \quad (3)$$

where  $\Delta \ln pop_{c,t}$  represents the change in CZ's  $c$  log population between years  $t$  and  $t-10$ :  $\Delta \ln(pop_{c,t}) = \ln(pop_{c,t}) - \ln(pop_{c,t-10})$ . We first examine changes in log population between 1990 and 2000 to capture pre-epidemic differential trends across CZs. We then analyze changes between 2010 and 2000, and between 2020 and 2000. To capture long-

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<sup>12</sup>To see this, consider that population changes could be model as an AR(1) process. That is,  $\Delta \ln(pop_{c,t}) = \theta \Delta \ln(pop_{c,t-1}) + \mu_{ct}$ . Not including the lagged population change when estimating the effects of exposure to the epidemic may mistakenly attribute effects of prior growth to this shock. Thus, the estimates of  $\phi$  from the following regression may be biased:  $\Delta \ln(pop_{c,t}) = \phi_t CancerMR_{c,1996} + \alpha X_{c,1990} + \gamma_s + \epsilon_{ct}$

term effects, we focus on 20-year changes in population, which helps avoid the inclusion of shorter-run lags already influenced by the epidemic. Specifically, when we define the outcome as the change in log population from 2020 to 2000,  $\Delta \ln(pop_{c,2020})$ , the relevant lag is the 20-year lag,  $\Delta \ln(pop_{c,2000})$ . We control for a vector of demographic characteristics  $X_{c,1990}$  and state-level fixed effects. We employ decadal U.S. Census data on CZ population counts to avoid using intercensal estimates.

Estimation of effects on sex-ratios follow a similar logic with the caveat that sex ratios *in levels* are autoregressive processes of order 1 (AR(1)), thus when estimating changes in sex ratios we control for the lagged level as follows:

$$\Delta \text{sex ratio}_{c,t} = \phi \text{CancerMR}_{c,1996} + \beta \text{sex ratio}_{c,t-10} + \alpha X_{c,1980} + \gamma_s + \epsilon_{ct}, \quad (4)$$

where  $\Delta \text{sex ratio}_{c,t} = \text{sex ratio}_{c,t} - \text{sex ratio}_{c,t-10}$  and the rest of the variables are defined as above. Appendix C. provides additional details on this approach.

**Identifying assumptions.** The key identifying assumption is that, in the absence of the opioid epidemic, the outcomes of areas with higher cancer mortality would have followed the same *trends* as those of areas with lower cancer mortality. We assess the plausibility of this assumption by examining pretrends in our event study analyses and by conducting additional robustness checks as described in Section VII..

Nonetheless, in designs with continuous exposures such as ours, and in the presence of treatment effect heterogeneity, this standard parallel trends assumption is not sufficient for identification. As shown by Goldsmith-Pinkham et al. (2020) and Callaway et al. (2024), stronger assumptions are required. Specifically, cancer mortality rates in 1996 must be exogenous not only to the evolution of the outcome variables but also to the magnitude of the treatment effects. In addition, treatment effect heterogeneity must be restricted such that observed differences across levels of cancer mortality reflect what would have occurred for all other “dose” groups had they been exposed to that same level. To assess the extent of treatment effect heterogeneity across “dose” levels, Goldsmith-Pinkham et al. (2020) recommend plotting first-stage and reduced-form estimates for subsamples along the distribution of the continuous exposure. Following this approach, we divide the sample into 15 equally sized groups based on 1996 cancer mortality rates, allowing for a finer-grained analysis across the full support of the exposure variable. Figure XXX plots the average cancer mortality rate in 1996 on the  $x$ -axis and the corresponding values of our outcome variables on the  $y$ -axis. This visualization transparently illustrates the reduced-form relationships between baseline cancer mortality and our various outcomes.

## V. Local Population and Local Labor Markets

While the opioid epidemic stands as one of the most tragic public health crisis of the past century—marked by excess drug-related mortality in the communities most exposed—it also induced significant deterioration in the economic fabric of these communities (see [Arteaga and Barone, 2025](#)). In this section, we first study the impacts of the epidemic on local population growth. These findings are of independent interest and have key implications for estimates of the labor market effects of the epidemic, to which we turn our attention in the second part of this section.

### V.a Population Changes

First, we document evidence that the local communities most exposed to the opioid epidemic experienced reduced population growth. Table 2 presents these results.

### V.b Economic Effects

Changes in the population levels and composition of places alters the composition of local labor markets and thus may confound estimates of the direct impact of the epidemic on local economic conditions. To see this, note that changes in the employment-to-population ratio can be decomposed as follows:

$$\Delta \frac{emp}{pop}_t = \frac{emp_t}{pop_t} - \frac{emp_0}{pop_0} = \underbrace{\frac{emp_t - emp_0}{pop_t}}_{\text{net emp change}} + \underbrace{\frac{emp_0}{pop_t} - \frac{emp_0}{pop_0}}_{\text{effect of pop change on emp/pop}} \quad (5)$$

In Figure 2, we find no detectable effects of the opioid epidemic on employment rates for men.<sup>13</sup> For women, we estimate negative effects, though they are not statistically significant. However, when we split the data by education, we observe that among women without a college education, the opioid epidemic induced a continuous decline in employment. By 2018, a one standard deviation increase in exposure to the epidemic led to an 8% reduction in employment for this group. We estimate a smaller impact of the opioid epidemic on men without college education, a one standard deviation increase in exposure to the epidemic lead to a 3% reduction in employment for this group.

Given these extensive margin changes, we ask how the earnings of these populations have changed. We find no detectable earning (aggregate) effects for women or men. When we zoom into education groups, we find changes that are consistent with women with some college education facing lower average earnings and displacing less educated

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<sup>13</sup>We need to update the employment results to define the outcome of interest as the “net employment change”. In this version, the outcome is defined as the left-hand side of Equation 5.

women out of the labor force. We find that the epidemic did not affected men earning significantly.

Panel (a) of Figure A1 provides evidence that prior to the onset of the opioid epidemic, receipt rates evolved similarly across communities with different baseline cancer mortality. Following the epidemic’s emergence, 1996 cancer mortality predicts a continuous increase in SNAP takeup in the subsequent two decades. In panel (b) we leverage data on the sex of applicants to document that this increased takeup was mainly driven by women.

## VI. Margins of Adjustment: Deaths, Moves, and Births

Having established the impacts on local population and economic outcomes, we now turn to potential explanations for these changes. We organize the discussion around three margins of adjustment. First, we explore a direct channel: deaths. Second, we examine migration responses to the epidemic. Finally, we consider the role of fertility.

### VI.a Deaths

The U.S. is now in its fourth decade of rising drug poisoning death rates and there is no end in sight with drug deaths exceeding 100,000 per year since 2021 (Evans and Lieber, 2025). Panel (a) and (c) of Figure 3 show the evolution of drug-induced mortality rates for men and women and for individuals without and with college in high and low cancer commuting zones. Men and those without college education experienced higher rates of drug-induced mortality. In 2018, the drug-related mortality rate for men was 2.5 times higher than that for women. However, when examining the effects of exposure to the opioid epidemic, we find that the effect sizes are broadly similar across sexes (panel (b) of Figure 3). Looking into effects by educational attainment, we find that the effects of exposure to the epidemic are not statistically different from zero for those with some college education but positive for those without college education. By 2010, a one-standard-deviation increase in our measure of exposure led to a 57% increase in the drug-induced mortality rate of this group.

**Counterfactual exercise.** Are these mortality impacts large enough to explain the population decline? To answer this question we consider a simple counterfactual exercise. We construct population counts for commuting zone  $c$  in year  $t$  under the assumption that mortality from drug-induced causes is null. To do so, we add the deaths of individuals who would be in the age bracket of interest to the observed population counts. Considering the population aged between 18 to 44 years old as an example, the counterfactual population count would be given by:

$$\widetilde{\text{pop}}_{ct,18-44} = \text{pop}_{ct,18-44} + \text{drug-induced deaths}_{c,[18-44]_t} \quad (6)$$

where drug-induced deaths <sub>$c,[18-44]_t$</sub>  is the sum of deaths from drug-induced causes among people who would be between 18 and 44 by year  $t$ .<sup>14</sup>

Figure A2 shows observed population trends and the counterfactual counts. There is little evidence that deaths could have played an important role in population decline among areas more exposed to the epidemic.

## VI.b Moves

Migration is consider the primary means of adjustment to local labor employment shocks (Blanchard et al., 1992, among others).

We then estimate the impact of the opioid epidemic on origin-destination migration flows by estimating the following regression for each annual cross-section of commuting zone pairs in year  $t$ :

$$\Delta y_{ol,t} = \phi_{o,t} Cancer_{o96} + \phi_{l,t} Cancer_{l96} + \alpha_{o,t} X_{o96} + \alpha_{l,t} X_{l96} + \gamma_{so} + \gamma_{sl} + v_{ol,t} . \quad (7)$$

Where  $y_{ol,t}$  is the log of the probability of migrating from commuting zone  $o$  to  $l$  in year  $t$ , and  $\Delta y_{ol,t}$  is the change in this variable between year  $t$  and the baseline year ( $t_0$ ).  $Cancer_{o96}$  and  $Cancer_{l96}$  capture exposure to the epidemic in the origin and destination locations, respectively.  $\gamma_{so}$  and  $\gamma_{sl}$  represent origin and destination state fixed effects. We include demographic controls in both origin and destination commuting zones at baseline ( $X_{o96}$  and  $X_{l96}$ ). This specification follows Borusyak et al. (2022) who suggest including exposure to shocks in origin and destination locations. Intuitively, this captures that moving decisions depend upon both the shock in the current location and the shock in the potential alternative locations.

The main coefficient of interest,  $\phi_{o,t}$ , captures the migration semi-elasticity with respect to exposure to the opioid epidemic. Specifically, holding all else constant, a one-unit increase in cancer mortality in the origin location leads to a  $\phi_{o,t}$  percent change in the probability of out-migrating from  $o$  by year  $t$ , relative to the baseline year. To ease the interpretation, we present scaled coefficients, i.e., we multiply the semi-elasticity by the standard deviation of cancer mortality in 1996.<sup>15</sup>

Figure 4 presents our results. Exposure to the opioid epidemic significantly increases out-migration rates. Panels (a) and (b) present complementary evidence on this pattern. Panel (a) presents estimates of inter-county out-migration effects using IRS data and

<sup>14</sup>In this computation we accumulated deaths from 1989, the first year of the mortality data. While this approach could introduce truncation concerns, this issue is mitigated by the fact that deaths from these causes began to rise only after 2000.

<sup>15</sup>Our use of cross-sectional regressions in our migration analysis is motivated by computational efficiency. It can be readily shown that estimating Equation 7 over all cross-sections is numerically equivalent to a stacked “event-study style” panel regression, analogous to Equation 1. We therefore continue to present our results in event-study style graphs and maintain the convention of indicating the baseline year with a zero coefficient.



shows that a one-standard deviation increase in exposure to the epidemic led to a 4% increase in out-migration rates.<sup>16</sup> This estimate reflects moves among individuals of all ages. Panel (b) presents estimates of inter-CZ migration focusing on individuals aged 15 to 44, who are of childbearing age and therefore are our primary sample of interest. In this sample, out-migration effects are more pronounced: a one-standard deviation increase in exposure led to a 12% increase in out-migration rates among women with no detectable effects among men.

We document significant heterogeneity in out-migration responses by sex and educational attainment. Panel (c) shows that inter-CZ out-migration responses are driven by women with some college education. This pattern of differential out-migration has important implications for fertility, given that educational attainment is a strong predictor of fertility behavior. For instance, using data from the ACS, we find that women without any college education have fertility rates by age 30, that are 48% higher than those of women with some college education.

Across all samples, we detect no effect of exposure to the opioid epidemic on in-migration rates. As a result, the net-migratory effects of the crisis are solely driven by changes in out-migration.

**Magnitudes and back of the envelope computations.** Are these migration responses large enough to account for the observed population decline? To address this question, we conduct a back-of-the-envelope exercise suggesting that migration responses were the primary driver of population changes between 2000 and 2010. Over the longer horizon, population changes between 2000 and 2020 can also be largely attributed to these migration dynamics.

**Where Do People Move?** We leverage detailed origin-to-destination migration data from the IRS to document how people move in response to the opioid epidemic. In particular, we aim to assess whether migrants relocate to better places—that is, whether their destinations exhibit lower exposure to the epidemic than their origins. To do so, we proceed as follows. We divide the data into 22 equally sized groups ( $g$ ) based on the origin’s exposure to the epidemic (i.e., 1996 cancer mortality).<sup>17</sup> For each group, we compute the weighted average exposure in the destination, where the weights ( $w_{ol}$ ) correspond to the share of movers from origin  $o$  who relocate to destination

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<sup>16</sup>The IRS migration data suppresses county-to-county migration flows below certain thresholds for confidentiality purposes, as a result, differential missingness across CZs makes aggregating to CZ-to-CZ migration flows challenging. At this paper’s current stage, when using the IRS migration data we focus on inter-county migration among county-pairs with 1996 migration rates in the top two quintiles which are unaffected by this suppression.

<sup>17</sup>We choose the number of groups that optimally fits the data.

$l(m_{ol})$ :

$$\overline{\text{Exposure}}_g^{dest} = \sum_{o \in g} \sum_d w_{ol} \text{Exposure}_l \quad \text{where} \quad w_{ol} = \frac{m_{ol}}{\sum_{l'} m_{ol'}}$$

Figure 5 graphically presents this measure for the 2010 moving patterns. A slope of one in the relationship between origin and destination exposure would indicate that individuals relocate to areas that are, on average, equally affected by the shock. We refer to this case as “no directness”. If the migratory response was directed from heavily-exposed areas to less-exposed ones, the slope would be negative. In the case where migration flows are perfectly directed—moving exclusively from the most to the least exposed areas—the slope would approach -1.

We find that movers targeted on average better places but this targeting was not perfect: individuals from more exposed areas tend to move to somewhat less exposed destinations, but not to the least exposed ones.

## VI.c Births

While the connection between population declined and migration is clear and strong, differential fertility patterns could provide a demographic tailwind allowing places to grow despite having negative net migration rates. In fact, [Asquith and Mast \(2024\)](#) argue that falling fertility has caused migration rates that used to generate growth to instead result in decline. In this section, we examine how exposure to the opioid epidemic affected fertility patterns across U.S. commuting zones.

We start by presenting the evolution of the fertility rate in areas more exposed to the epidemic—those in the top quartile of cancer mortality in 1996—compared to areas with less exposure—the bottom quartile of 1996 cancer mortality. Panel (a) of Figure 6 shows that while fertility rates have generally declined over the past three decades, the decline was less pronounced in areas more exposed to the epidemic. Estimating Equation (1), we find that exposure to the opioid epidemic led to a substantial (relative) increase in fertility among women of reproductive age. Panel (b) of Figure 6 shows that a one standard deviation increase in exposure—proxied by local 1996 cancer mortality rates—led to an approximately 5.5% increase in the overall fertility rate among women aged 15 to 44 by 2018, relative to its 1996 baseline. These effects start to be significant in 2005, and have steadily grown since then. These relative increases in fertility are driven by women aged 25 to 29. Figure A3 presents a breakdown of the fertility results by age and shows that this is the only group in which we observe a significant effect of exposure to the epidemic.

Panel (a) of Figure A4 presents the effects of exposure on fertility by marital status. We find that a one-standard deviation increase in exposure raised fertility among un-

married women by 4.7% by 2018, compared to a 2.8% increase among married women. The dynamics of these effects also differ meaningfully: the increase in fertility among unmarried women becomes statistically significant starting in 2005, whereas it takes an additional eight years—until 2013—for the effect on married women’s fertility to reach statistical significance.

When we disaggregate the effects by education level, we find that the increase in fertility is more pronounced among women without any college education, particularly during the decade from 2005 to 2015. Panel (b) of Figure A4 shows that in areas more exposed to the opioid epidemic, fertility among women without a college degree increased by 5% by 2010. In contrast, the increase among women with some college education was only 1.8% over the same period. Over time, fertility rates for both groups continued to rise. By 2018, fertility had increased by 6% among women without any college education and by 5% among those with some college or more.

These effects on fertility rates have two important implications. First, in terms of population declined, the direction of these changes indicate that (relative) higher fertility induced by the epidemic could attenuate its impacts on population growth. Second, the heterogeneous fertility changes will contribute to long term changes in population composition in areas more affected by the epidemic.

## VII. Robustness Checks

In this section, we provide evidence in support of our identification strategy, explore alternative explanations for our findings and test the sensitivity of our results.

### VII.a Evidence on the Exogeneity Assumption and Exclusion Restriction

The validity of the identification strategy requires that, in the absence of prescription opioid marketing, areas with higher cancer mortality in 1996 would have followed the same trends in health and economic outcomes as areas with lower cancer mortality—all along the support of the 1996 cancer mortality distribution. To support this assumption, we present several exercises.<sup>18</sup>

#### Out-of-sample.

**Placebo.** We further probe the validity of our design by estimating event-study regressions with placebo instruments—i.e., mid-1990s mortality from causes unrelated to cancer.<sup>19</sup> Finding a good placebo instrument is challenging given that the causes that underlie the incidence of cancer and of other conditions such as heart disease are not

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<sup>18</sup>In this draft we focus on population growth as the main outcome of analysis. Arteaga and Barone (2025) provide complementary evidence on the validity of this strategy and present results for these tests on mortality, economic, and political outcomes.

<sup>19</sup>These placebo instruments are also known as negative instruments (Danieli et al., 2023).

independent (Chiang, 1991; Honoré and Lleras-Muney, 2006). As a result, there is substantial overlap across underlying causes, and the correlation across measures is very high, especially among elderly age groups. *Discuss results*

**Additional covariates.** We also expand the set of covariates to account for health behaviors, access to healthcare and economic factors and the correlation of these with cancer. For example, Dong et al. (2022) document that the share of smokers and the share of adults with overweight are among the main risk factors associated with cancer mortality. First, we show that including contemporaneous cancer mortality in our set of controls also does not alter our results (*add*). We then expand the set of controls in our baseline to include the share of smokers, the share of adults with overweight, the share of primary care physicians, and the infant mortality rate, all measured at the CZ level at baseline or the earliest available data point. *Discuss results*

Similarly, we include a set of economic covariates to our main specification. We interact each economic control measured at baseline with year dummies; these controls include the unemployment rate, the share of employment in the manufacturing sector, income per capita, and the share of the population that has completed some college.<sup>20</sup> *Discuss results*

## VII.b Continuous Exposure and Treatment Effect Heterogeneity

### VII.c Exposure to Economic Shocks

We account for geographic exposure to major economic shocks that could confound the relationship between opioid exposure and our outcomes. For each shock, we include a measure of local exposure interacted with year dummies to flexibly control for its varying geographic impact.<sup>21</sup>

Contemporaneous with the unfolding of the opioid epidemic, the United States economy was affected by rising import competition from China and two major recessions (2001 and 2007–09). These developments could influence the same family formation and economic outcomes we study. To address rising import competition from China, we control for exposure to the regional import competition measure from Pierce and Schott (2020). Exposure to the 2001 and Great Recession is measured using changes in CZ-level unemployment around each downturn, as in Yagan (2019). *Discuss results*

### VII.d Alternative Sample Restrictions

Finally, we assess whether our results are sensitive to the exclusion of any single state or driven by specific regions of the country. *Discuss results*

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<sup>20</sup>All of these controls are measured at the CZ level at baseline or the first available year.

<sup>21</sup>We follow the literature to construct these measures; (Arteaga and Barone, 2025) provides the details. Correlations with opioid exposure range from  $-0.04$  to  $0.18$ : NAFTA:  $-0.04$ , China shock:  $0.18$ , 2001 recession:  $0.05$ , Great Recession:  $0.02$ .

## VIII. Discussion

This paper documents how the opioid epidemic, one of the most consequential public health crises in recent U.S. history, shaped two fundamental demographic processes: fertility and internal migration. Using exogenous variation in exposure to the epidemic stemming from Purdue Pharma’s early marketing strategy for OxyContin, we show that more heavily exposed communities experienced a substantial rise in fertility—particularly among younger, unmarried, and less-educated women. At the same time, we find strong evidence of selective out-migration, concentrated among college-educated women and those with lower expected early-life fertility. These compositional shifts—rather than changes in fertility preferences alone—appear to be a key mechanism behind the rise in fertility in exposed areas.

Our findings carry several broader implications. First, they reveal a novel channel through which public health crises can influence long-term demographic and social patterns—not only through direct health effects but also via changes in family formation and population structure. Second, they underscore the importance of considering both behavioral and compositional responses when evaluating the demographic consequences of large-scale shocks. Finally, by showing how the opioid epidemic has contributed to demographic divergence across communities, our results speak to the growing geographic inequality in social and economic conditions within the United States.

Future research could build on our findings by examining how these shifts in family formation and migration interact with other long-run outcomes, such as child development, educational attainment, or intergenerational mobility. Additionally, understanding how institutional responses—such as access to addiction treatment, economic support programs, or housing policy—moderate these effects remains an important avenue for further exploration.

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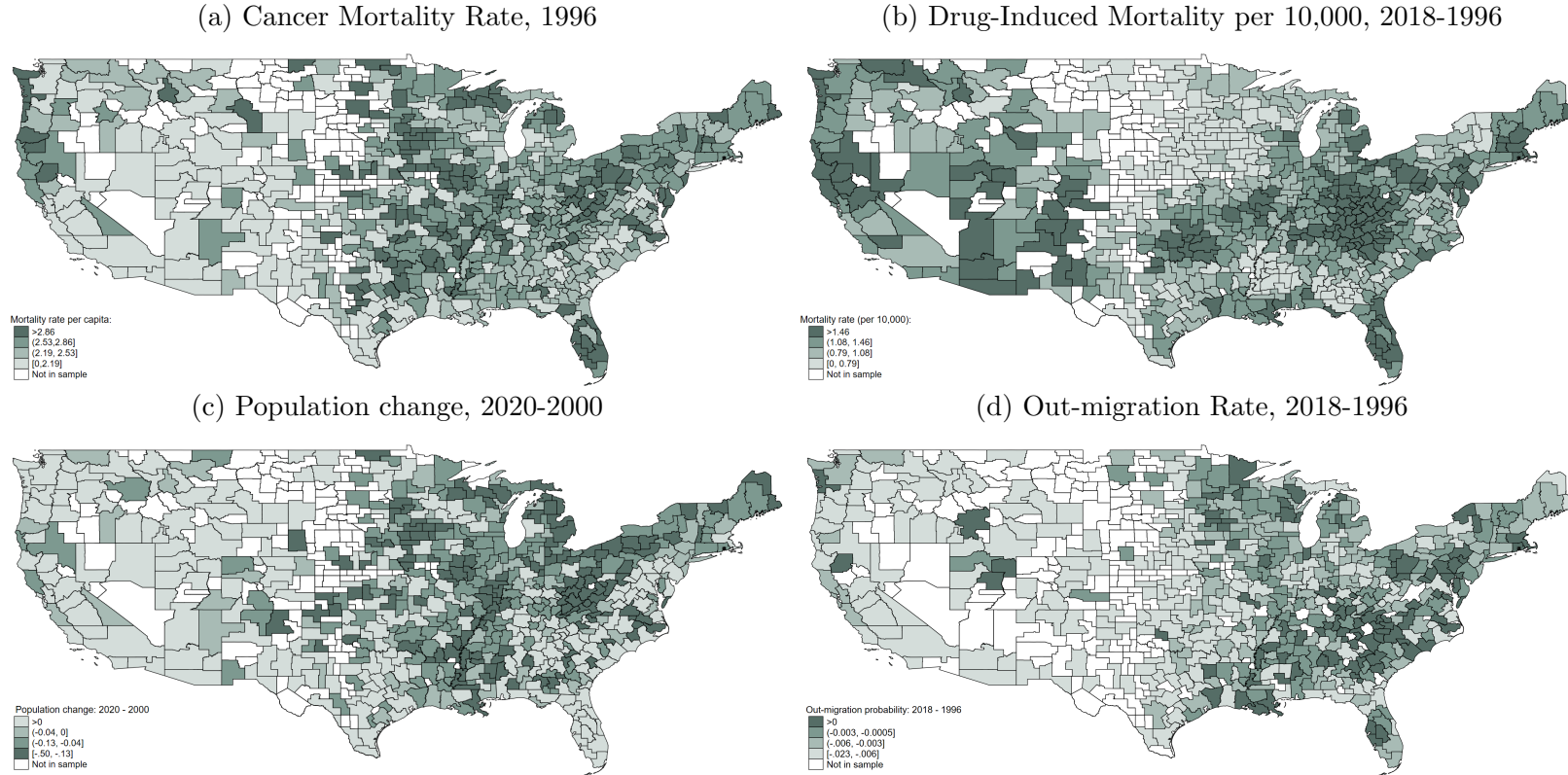
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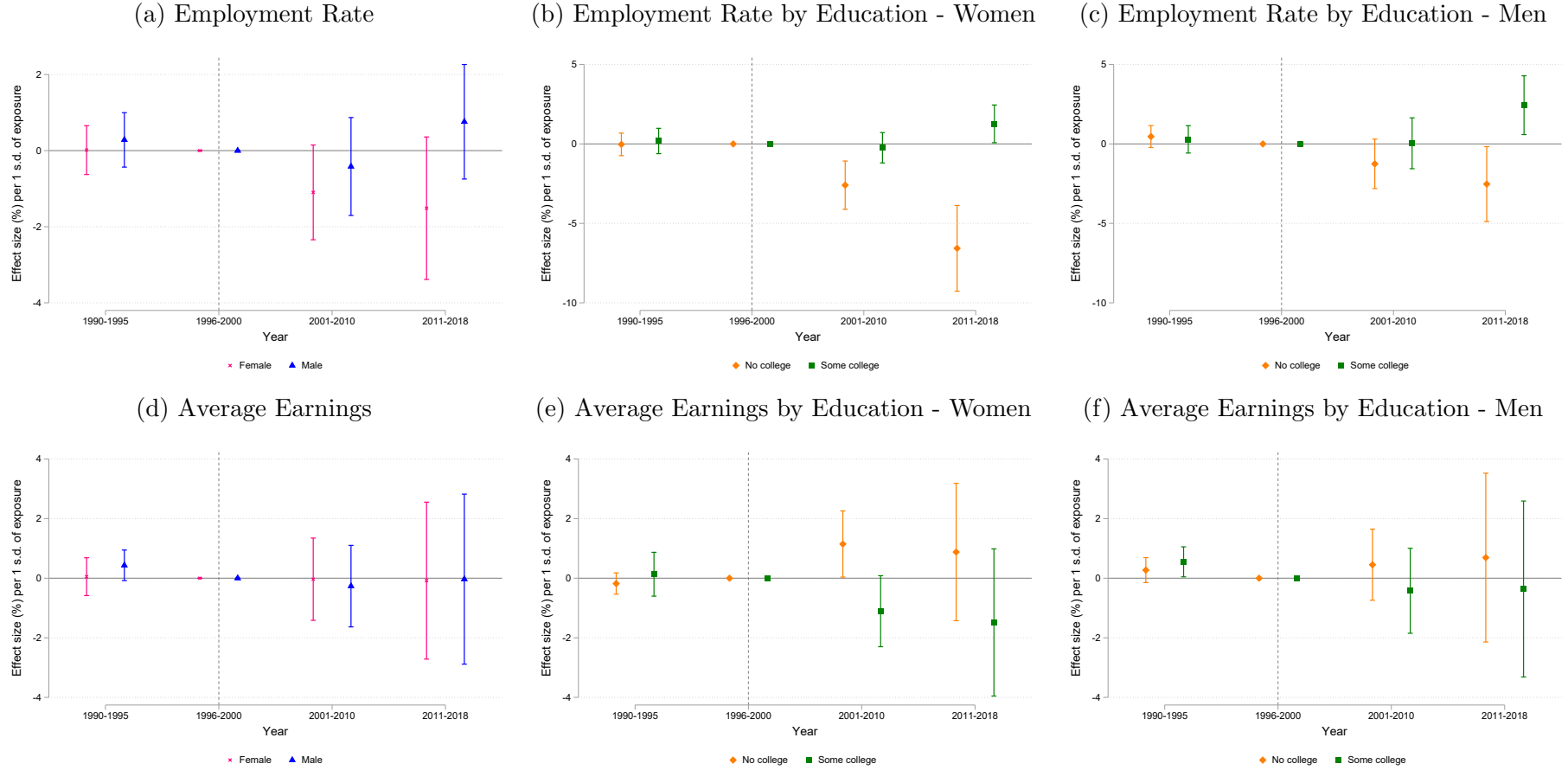
## IX. Figures

Figure 1: Geographic Variation



Notes: Panel (a) of this figure shows the geographic distribution of the cancer mortality rate (per 1,000) in 1996, our measure of exposure to the opioid epidemic. Panel (b) shows the geographic distribution of drug-induced mortality (per 10,000) between 1996 and 2018. Panel (c) shows the change in the (natural log of) population aged 18 to 44 years old between the decennial years 2020 and 2000. Panel (d) shows the change in out migration probabilities from 1996 to 2018. The break points correspond to the quartiles of the variable's distribution. In the bottom panels, we use zero as a break point to ease interpretation of population decline (panel c) and net-out migration (panel d). The sample is restricted to areas with more than 25,000 residents in 1996; these account for more than 97% of the total population in 1996. Commuting zones not included in the sample are white in the figure. We use geocoded National Vital Statistics System (NVSS) data to construct death counts; see Section III. for the ICD codes used. This figure is referenced in Section III.

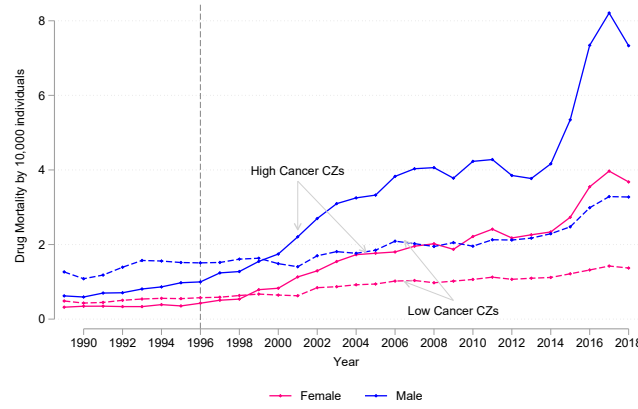
Figure 2: Exposure to the Opioid Epidemic and Employment Rates and Earnings by Sex and Education



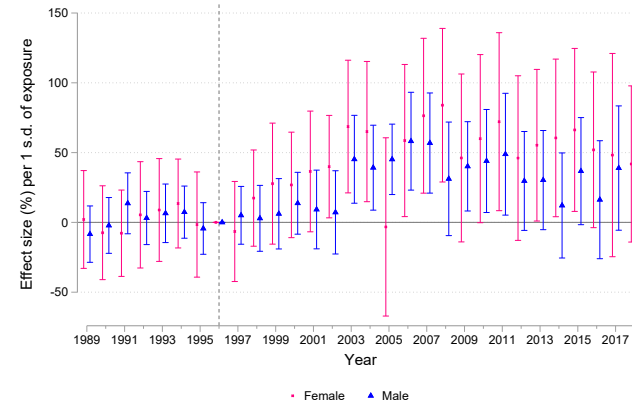
Notes: This figure presents the effects of exposure to the opioid epidemic on employment rates and average earnings by sex and sex by education. We estimate Equation (2). We present scaled  $\phi_g$  coefficients and referred to these as the effect size (expressed as a percentage). Specifically, we multiply the estimated coefficients by the standard deviation of 1996 cancer mortality rates and divide by the mean of the outcome variable in 1996. These estimates exploit continuous variation in cancer mortality in 1996 and control for baseline characteristics and state-by-year fixed effects. Standard errors are clustered at the CZ level. This figure is referenced in Section V.b

Figure 3: Exposure to the Opioid Epidemic and Drug Induced Mortality by Sex and Education

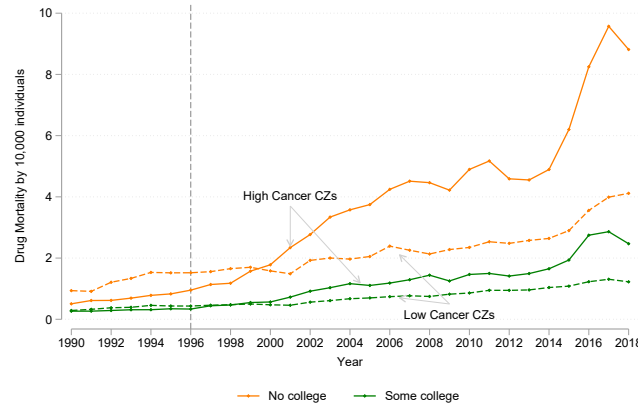
(a) Trends in Drug Induced Mortality by Sex



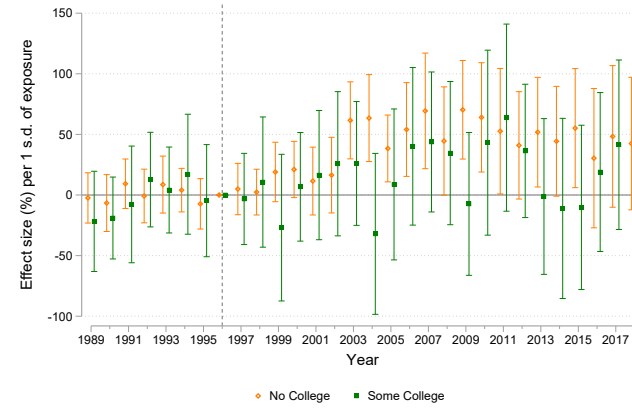
(b) Event-study by Sex



(c) Trends in Drug Induced Mortality by Education

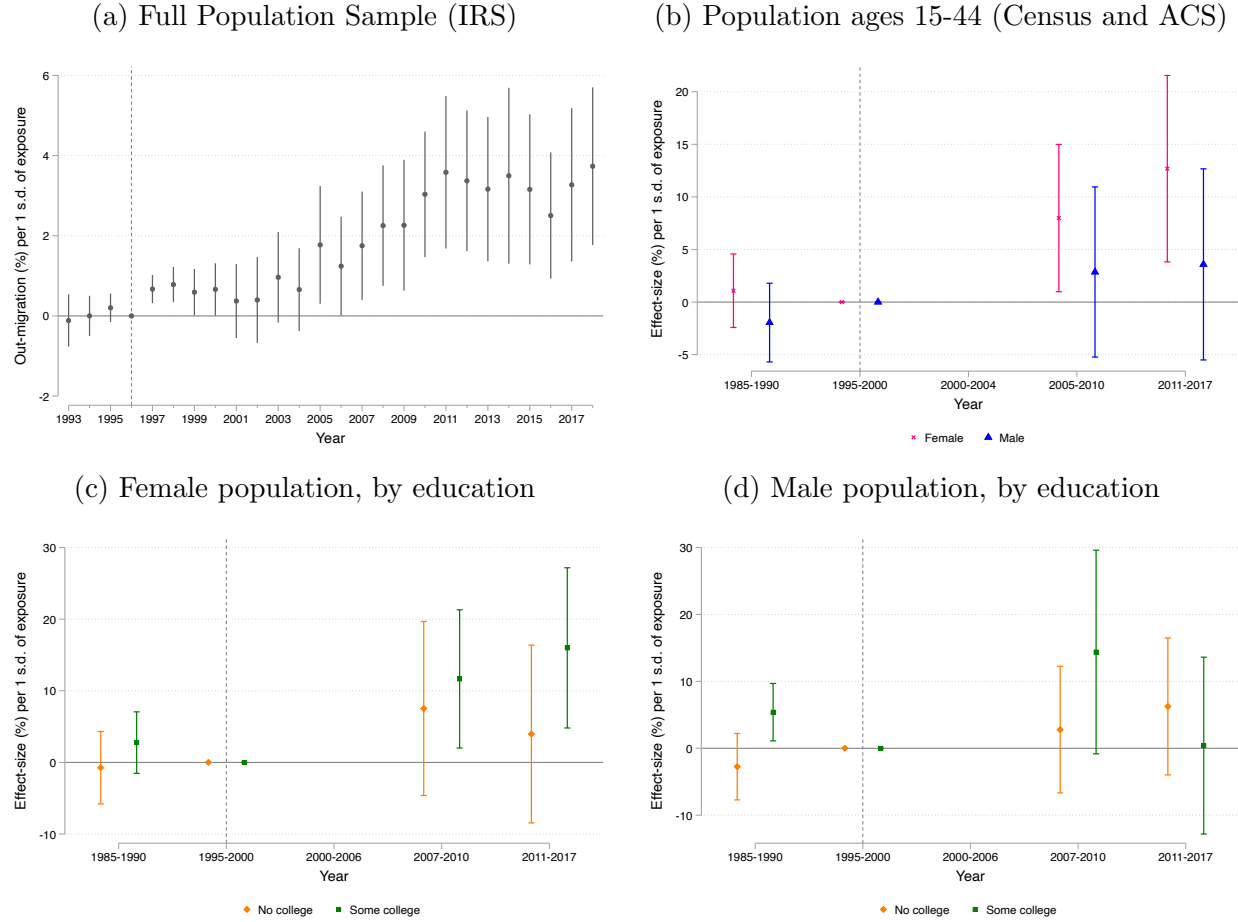


(d) Event-study by Education



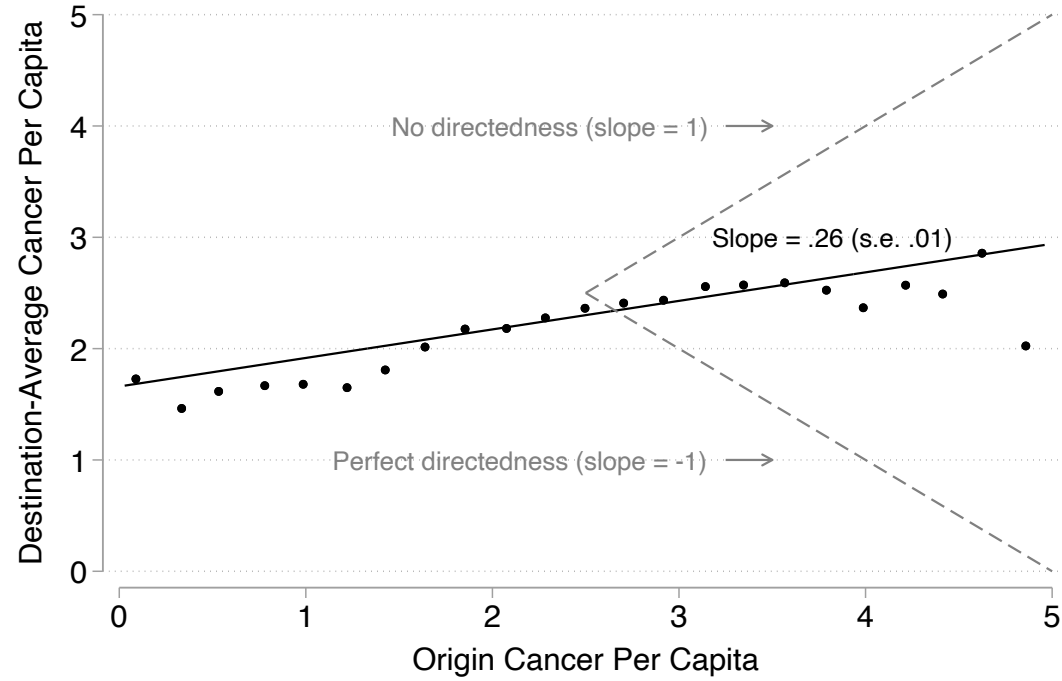
Notes: This figure shows the effects of the opioid epidemic on drug-induced mortality of women and men aged 20 to 44 years old. The effect size (expressed as a percentage) corresponds to the scaled coefficients from Equation (1). Specifically, we multiply the estimated coefficient by the standard deviation of 1996 cancer mortality rates and divide by the mean of the outcome variable in 1996. These estimates exploit continuous variation in cancer mortality in 1996 and control for baseline characteristics and state-by-year fixed effects. Standard errors are clustered at the CZ level. This figure is referenced in Section ??

Figure 4: Effects of the Opioid Epidemic on Out-Migration



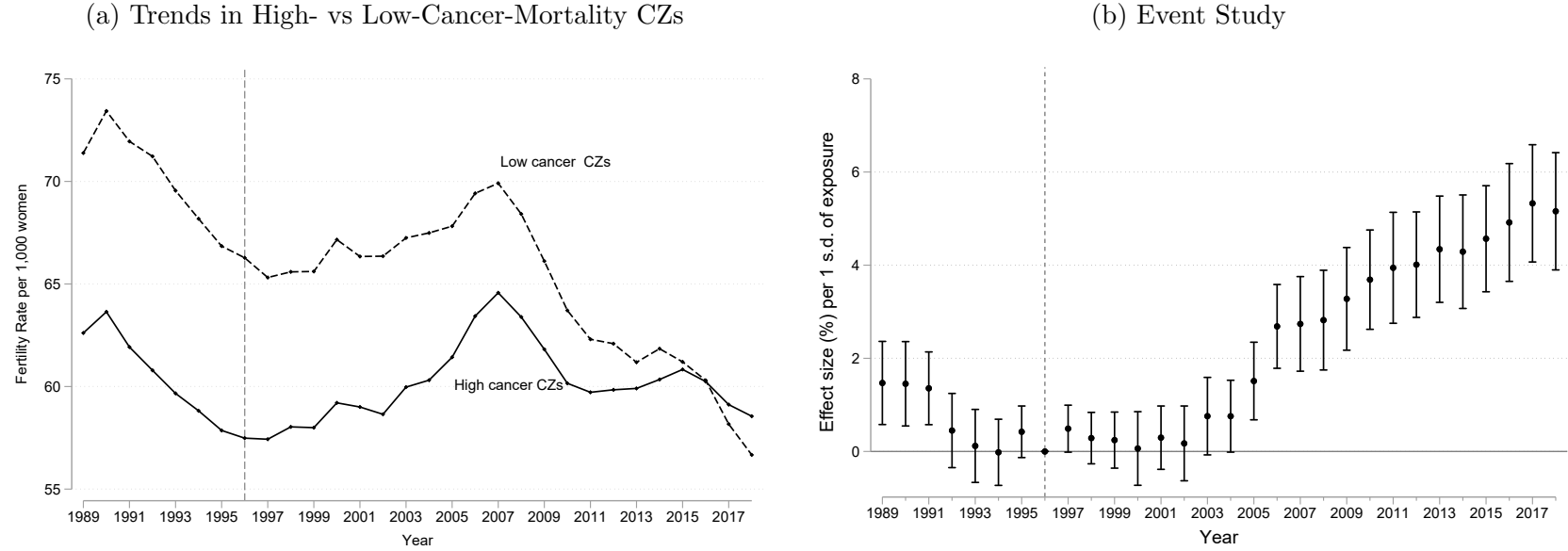
Notes: This figure presents the effects of exposure to the opioid epidemic on migration rates. Data from the IRS (panel a) covers inter-county internal migration for individuals of all ages and provides year to year moves. The IRS suppresses county-to-county migration flows below certain thresholds for confidentiality purposes; we therefore focus our analysis on county-pairs with 1996 migration rates in the top two quintiles. Data from the Census and ACS (panels b to d) are used to construct inter-CZ migration of individuals aged 15 to 44. Data coverage restricts the analysis to years 1990, 2000, and 2007 onward. The effect size (expressed as a percentage) corresponds to the scaled coefficients  $\phi_o$  from Equation (7). Specifically, we multiply the estimated coefficient by the standard deviation of 1996 cancer mortality rates. Standard errors are clustered two-way at the origin-destination level. This figure is referenced in Section VI.b

Figure 5: Where Do People Move?



Notes: This figure plots the average exposure in destinations against the exposure level in origin county, where exposure is measured by cancer mortality per capita in 1996. Origins are grouped into 22 equally sized bins based on exposure. For each group, we compute the (weighted) average exposure in destinations, where weights correspond to the share of movers from origin  $o$  who relocate to destination  $l$ . The dashed lines represent reference slopes of 1 (no directedness) and  $-1$  (perfect directedness). This figure is referenced in Section XX .

Figure 6: Effects of the Opioid Epidemic on Total Fertility



Notes: Total fertility is defined as the ratio of births to women aged 15 to 44 to the female population in this age group. Panel (a) shows the evolution of the fertility rate per 1,000 women in commuting zones (CZs) in the bottom (dashed lines) and top (solid lines) quartiles of cancer mortality prior to the launch of OxyContin. This comparison is based on a binary high-versus-low classification of CZs, with the outcome weighted by population. Panel (b) presents estimates of the effect of exposure to the opioid epidemic on fertility rates. The effect size (expressed as a percentage) corresponds to the scaled coefficients from Equation (1). Specifically, we multiply the estimated coefficient by the standard deviation of 1996 cancer mortality rates and divide by the mean of the outcome variable in 1996. These estimates exploit continuous variation in cancer mortality in 1996 and control for baseline characteristics and state-by-year fixed effects. This figure is referenced in Section VI.c.



## X. Tables

Table 1: Summary Statistics

|                                                              | 1989-1996 |          |         | 2010-2018 |          |          |
|--------------------------------------------------------------|-----------|----------|---------|-----------|----------|----------|
|                                                              | Mean      | Median   | SD      | Mean      | Median   | SD       |
|                                                              | (1)       | (2)      | (3)     | (4)       | (5)      | (6)      |
| <b><i>Drug-induced mortality per 10,000 (ages 20-44)</i></b> |           |          |         |           |          |          |
| All                                                          | 0.0506    | 0.0416   | 0.0501  | 0.1950    | 0.1567   | 0.1738   |
| Male                                                         | 0.0680    | 0.0503   | 0.0788  | 0.2474    | 0.1961   | 0.2372   |
| Female                                                       | 0.0332    | 0.0197   | 0.0453  | 0.1412    | 0.1123   | 0.1434   |
| <b><i>Cancer mortality rate per 1,000</i></b>                | 2.5218    | 2.5227   | 0.5477  | 2.4858    | 2.4998   | 0.5517   |
| <b><i>Fertility rate (per 1,000 women)</i></b>               |           |          |         |           |          |          |
| Total fertility rate (15-44 y.o.)                            | 63.6284   | 62.3441  | 10.1162 | 63.4227   | 62.5304  | 9.4574   |
| Women aged 20-24                                             | 128.1476  | 129.8867 | 35.7958 | 100.4293  | 100.2160 | 34.0146  |
| Women aged 25-29                                             | 114.1135  | 111.3543 | 18.7233 | 121.4003  | 118.4229 | 22.1958  |
| Women aged 30-34                                             | 64.7392   | 63.7197  | 14.5065 | 86.6902   | 85.9445  | 16.4032  |
| Unmarried Women (18-44 y.o.)                                 | 80.2559   | 77.9313  | 13.9817 | 84.9512   | 83.3644  | 12.9375  |
| Married Women (18-44 y.o.)                                   | 45.6562   | 43.7260  | 16.4329 | 55.5676   | 55.1938  | 15.8463  |
| Without college (18-44 y.o.)                                 | 86.8485   | 84.0352  | 18.8434 | 82.9502   | 83.8315  | 22.3694  |
| With college (18-44 y.o.)                                    | 51.5866   | 51.4032  | 10.7864 | 58.2836   | 59.2978  | 15.8951  |
| <b><i>Marriage rates</i></b>                                 |           |          |         |           |          |          |
| Share of married women (18-44 y.o.)                          | 0.6241    | 0.6302   | 0.0637  | 0.4732    | 0.4770   | 0.0671   |
| Without college                                              | 0.6379    | 0.6369   | 0.0611  | 0.3971    | 0.3971   | 0.0868   |
| With college                                                 | 0.6222    | 0.6241   | 0.0743  | 0.5091    | 0.5045   | 0.0849   |
| Share of married men (18-44 y.o.)                            | 0.5572    | 0.5628   | 0.0550  | 0.4014    | 0.4036   | 0.0568   |
| Without college                                              | 0.5283    | 0.5313   | 0.0539  | 0.3374    | 0.3326   | 0.0693   |
| With college                                                 | 0.6011    | 0.6110   | 0.0703  | 0.4646    | 0.4650   | 0.0834   |
| <b><i>Sex ratios</i></b>                                     |           |          |         |           |          |          |
| Men (18-54 y.o.) to women (18-44 y.o.)                       | 1.2916    | 1.2829   | 0.0929  | 1.4794    | 1.4529   | 0.1570   |
| Without college                                              | 1.1822    | 1.1734   | 0.1126  | 1.1362    | 1.1409   | 0.1468   |
| With college                                                 | 1.3513    | 1.3414   | 0.1237  | 1.8437    | 1.7906   | 0.2971   |
| <b><i>Economic outcomes</i></b>                              |           |          |         |           |          |          |
| Women employment to pop                                      | 0.5431    | 0.5333   | 0.0969  | 0.5852    | 0.5796   | 0.1029   |
| Women earnings                                               | 18811.47  | 18154.19 | 3113.33 | 34507.38  | 33395.86 | 5935.11  |
| Men employment to pop                                        | 0.5702    | 0.5634   | 0.1181  | 0.5748    | 0.5702   | 0.1246   |
| Men earnings                                                 | 31727.50  | 30842.23 | 5219.93 | 53411.69  | 51664.56 | 10343.84 |

Notes: This table presents summary statistics for the main dependent variables and our measure of exposure to the opioid epidemic for the periods before and after the launch of OxyContin. This table is referenced in Section III.

Table 2: Effects of Exposure to the Opioid Epidemic on Population Growth Rates

| 1996 cancer mortality $\times$ | Change in log population per 1 s.d. exposure |                        |                        |                        |
|--------------------------------|----------------------------------------------|------------------------|------------------------|------------------------|
|                                | All ages<br>(1)                              | Ages 0-17<br>(2)       | Ages 18-44<br>(3)      | Ages 45-64<br>(4)      |
| 1990 – 2000                    | -0.0015<br>(0.0056)                          | -0.0025<br>(0.0077)    | -0.0134<br>(0.0072)    | -0.0012<br>(0.0052)    |
| 2000 – 2010                    | -0.0123***<br>(0.0034)                       | -0.0183***<br>(0.0049) | -0.0247***<br>(0.0048) | -0.0220***<br>(0.0034) |
| 2000 – 2020                    | -0.0253**<br>(0.0090)                        | -0.0307**<br>(0.0104)  | -0.0416***<br>(0.0103) | -0.0524***<br>(0.0095) |
| Number of CZs                  | 587                                          | 587                    | 587                    | 587                    |

Notes: This table presents estimates of the effects of exposure to the opioid epidemic on the change in CZ log population. We consider changes in the total population and by age groups. Each cell reports estimates of the coefficient  $\phi$  from Equation (3). All specifications include a set of demographic controls measured in 1990 and state-level fixed effects. Standard errors are clustered at the CZ level. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\*  $p < 0.01$ . This table is referenced in Section XXX.

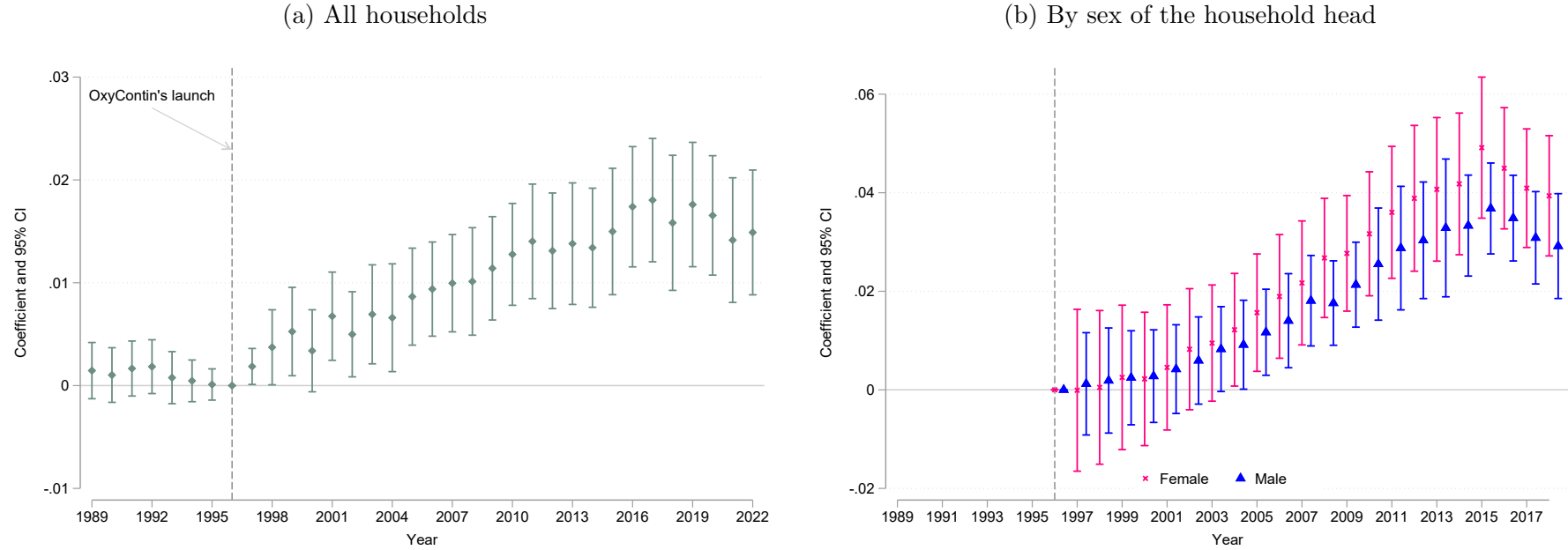
Table 3: Effects of Exposure to the Opioid Epidemic on Population Sex Ratios

| 1996 cancer mortality $\times$ | Change in sex-ratios per 1 s.d. exposure |                     |
|--------------------------------|------------------------------------------|---------------------|
|                                | Ages 18+<br>(1)                          | Ages 18-44<br>(2)   |
| 1980 – 1990                    | -0.0045***<br>(0.0013)                   | -0.0021<br>(0.0020) |
| 1990 – 2000                    | 0.0004<br>(0.0034)                       | 0.0017<br>(0.0058)  |
| 2000 – 2010                    | 0.0061***<br>(0.0017)                    | 0.0071*<br>(0.0032) |
| 2000 – 2020                    | 0.0034<br>(0.0023)                       | -0.0001<br>(0.0035) |
| Number of CZs                  | 587                                      | 586                 |

Notes: This table presents estimates of the effects of exposure to the opioid epidemic on the change in CZ sex-ratios. The sex ratio is the the number of men in an aged bracket over the number of women in the same age bracket. Each cell reports estimates of the coefficient  $\phi$  from Equation (4). All specifications include a set of demographic controls measured in 1990 and state-level fixed effects. Standard errors are clustered at the CZ level. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\*  $p < 0.01$ . This table is referenced in Section XXX.

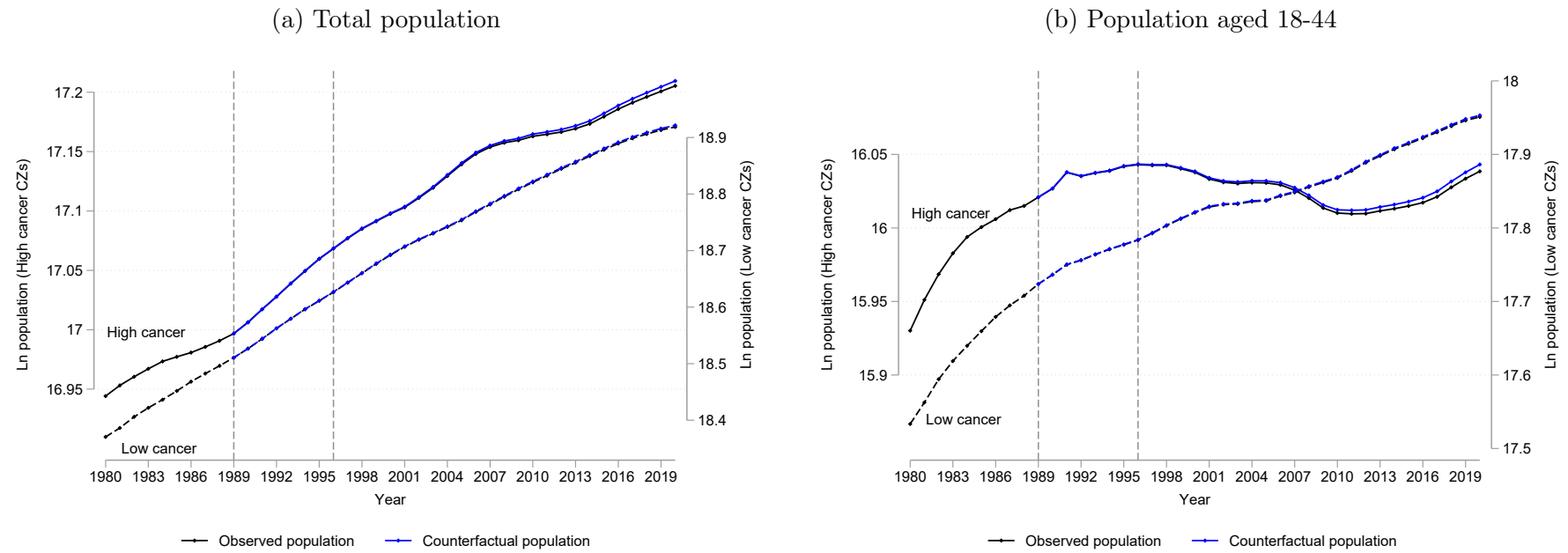
## A. Additional Figures and Tables

Figure A1: Effects of the Opioid Epidemic on SNAP Receipt Rate



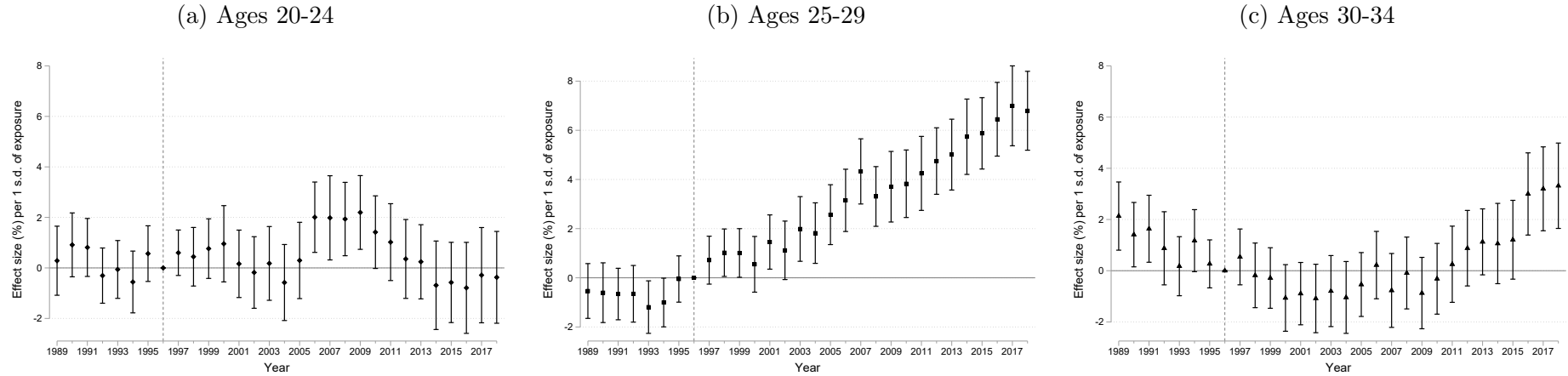
Notes: Panel (a) of this figure shows the effects of exposure to the opioid epidemic on SNAP receipt rates, which corresponds to the ratio between number of beneficiaries of the SNAP program and the population of the CZ. This event-study exploit continuous variation in 1996 cancer mortality and include time-varying controls and CZ and state-year fixed effects. Standard errors are clustered at the CZ level. Panel (b) of this figure shows the effects on SNAP receipt rates by sex of the head of the household participating in the program. Due to data limitations, we estimate these effects at the state level using the following specification:  $y_{st} = \sum_{\tau=1997}^{2018} \phi_{\tau} \text{CancerMR}_{c,1996} \mathbf{1}(\text{Year} = \tau) + X_{st} + \gamma_s + \gamma_t + v_{st}$ . Where the vector  $X_{st}$  includes the contemporaneous cancer mortality, the white and female population shares, and the shares of the population aged 18–29, 30–49, 50–64, above 65 years, and under 1 year. This figure is referenced in Section XXX.

Figure A2: Population Trends: Observed vs Counterfactual



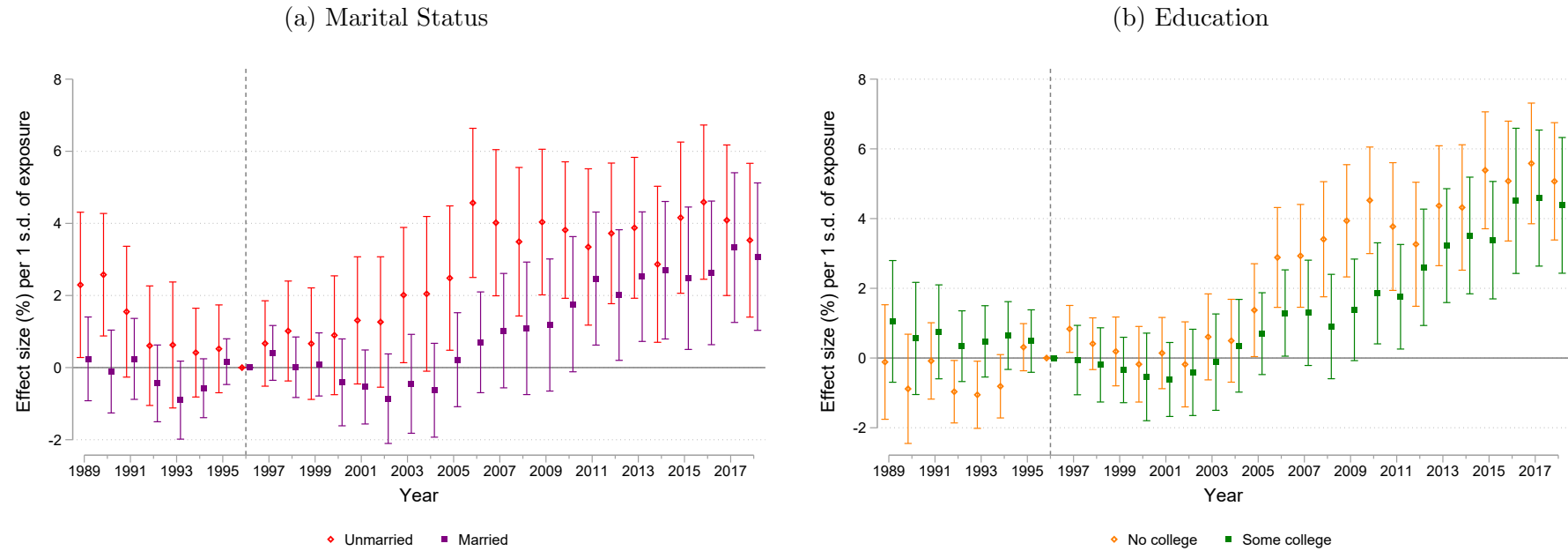
Notes: TBA. This figure is referenced in Section XXX .

Figure A3: Effects of the Opioid Epidemic on Fertility by Age Groups



Notes: This figure presents the effects of exposure to the opioid epidemic on fertility rates by age group. The three groups together accounted for 76% of births to women aged 15–44 in 1996: 25% were to mothers aged 20–24, 28% to those aged 25–29, and 23% to those aged 30–34. Figure ?? shows the evolution of these shares. The effect size (expressed as a percentage) corresponds to the scaled coefficients from Equation (1). Specifically, we multiply the estimated coefficient by the standard deviation of 1996 cancer mortality rates and divide by the mean of the outcome variable in 1996. These estimates exploit continuous variation in cancer mortality in 1996 and control for baseline characteristics and state-by-year fixed effects. Standard errors are clustered at the CZ level. This figure is referenced in Section VI.c

Figure A4: Effects of the Opioid Epidemic on Fertility by Marital Status and Education



Notes: This figure shows the effects of exposure to the opioid epidemic on fertility rates by marital status (panel a) and educational level of the mother (panel b). The effect size (expressed as a percentage) corresponds to the scaled coefficients from Equation (1). Specifically, we multiply the estimated coefficient by the standard deviation of 1996 cancer mortality rates and divide by the mean of the outcome variable in 1996. These estimates exploit continuous variation in cancer mortality in 1996 and control for baseline characteristics and state-by-year fixed effects. The married (unmarried) fertility rate is defined as the number of births to married (unmarried) individuals aged 15 to 44 years old divided by the population of married (unmarried) women in that age group. Similarly, fertility rates by education are the ratio of births to individuals with a no college (some college) to the number of women with no college (some college). Standard errors are clustered at the CZ level. This figure is referenced in Section VI.c.

## **B. Additional Empirical Evidence: 1996 Cancer Mortality Rates as a Measure of Exposure**

### **B.1 Pharmaceutical Marketing and Mid-Nineties Cancer**

1994 MS Contin prescription rates are strongly correlated with 1996-1998 OxyContin prescription rates and with 1996 cancer mortality rates

High-cancer CZs received a greater share of payments and visits from pharmaceutical companies to promote opioids

### **B.2 Cancer Mortality & Prescription Opioid Shipments**

Evidence suggesting that high-cancer CZs experienced a substantially larger increase in opioid prescriptions than low-cancer CZs

### **B.3 Robustness**

Expand the set of covariates to account for health behaviors, access to healthcare, and economic factors and the correlation of these with cancer; our results remain unchanged.

Include a set of economic covariates in our main specification, the inclusion of these covariates does not affect our main estimations.

Account for geographic exposure to major economic shocks that could confound the relationship between opioid exposure and our outcomes; controlling for these economic shocks does not alter our results.

## **C. Population Regressions**

In this appendix we provide more details on how to model population growth.